TOPIC 06

Multivariate Linear Regression

Regression with several independent variables: the model, how to read it, and why leaving a variable out can mislead

WHAT YOU WILL LEARN

- Explain why one predictor is often not enough
- Write the population and sample multiple regression equations
- Interpret a coefficient as an effect with the other variables held constant
- Describe the geometry for two or more predictors
- See omitted-variable bias in a worked example
- List the conditions needed for the model to be trusted

NOTE **Where this connects**

So far every model had a single regressor: $\hat{y} = b_0 + b_1 x$ (Topics 03 to 05). We found $R^2 = 0.41$ for college GPA on SAT score alone, and Topic 04 concluded that 'we are missing something'. Multivariate regression adds those other variables. Everything else (OLS, R^2, tables, t-tests) carries over.

1. Why one variable is not enough

Real outcomes have many causes.

Outcome (y)	Possible independent variables
House price	Size, location, year of construction
College GPA	SAT score, family income, gender, and more
Income	Household income, education, tenure, country, languages spoken

A good model needs to address the **higher complexity** of the problem, and the more factors you take into account, the better it can describe the outcome. Some real problems depend on more than 20 factors.

2. The model

Compare the two equations (the lecture slide 'Population model'):

$$y = \beta_0 + \beta_1 x_1 + \varepsilon \qquad \longrightarrow \qquad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

and the estimated version computed from a sample, the **multivariate regression equation**:

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$

Symbol	Meaning
$\hat{y}$	Predicted (inferred) value of the dependent variable
b_0	Intercept: $\hat{y}$ when all x are 0
$x_1, x_2, \ldots, x_k$	The k independent variables
$b_1, b_2, \ldots, b_k$	Their coefficients; b_j goes with x_j
β and ε	Population coefficients and the error term (what no x explains)

The only difference from simple regression is **more independent variables** instead of one. It is still *linear*: the equation is linear in the coefficients.

> **KEY IDEA** **How to read b_j**
> b_j is the expected change in y for a one-unit increase in x_j **while all the other variables are held constant**. Example: in a house-price model, the coefficient on size is the price difference between two houses of the *same age and location* that differ by one square metre.

Categorical variables such as gender

The lecture lists gender as a possible predictor of GPA. A category cannot be typed into the equation as text, so it is coded as a **dummy variable** that is 1 for one group and 0 for the other (for example female = 1, male = 0). Its coefficient is the average difference in y between the two groups, **holding the other variables constant**. A variable with more than two categories uses several dummies.

Geometry

Number of predictors	Fitted object
1	a line
2	a plane in 3-D (see Figure 1)
3 or more	a hyperplane (cannot be drawn)

With more than one predictor we can no longer speak of a 'best-fitting line'. OLS still works the same way: it picks the b values that minimise SSE.

3. A worked example with simulated data

Real multivariate data sets are large, so we simulate one. Suppose 120 houses have **size** (m^2), **age** (years) and **price** (thousands). Larger houses tend to be newer, which makes size and age correlated. The true rule is price = 50 + 2.0 × size − 1.5 × age + noise.

Model	b_0	b(size)	b(age)	R^2
Size only (simple regression)	−64.1	2.39	—	0.958
Size and age (multivariate)	62.9	1.96	−1.72	0.982
True rule used to simulate	50	2.00	−1.50	

- The multivariate fit comes close to the true effects: 1.96 per m^2 (true 2.00) and −1.72 per year of age (true −1.50).
- The size-only regression reports **2.39** per m^2, a different answer, because size and age move together. Bigger houses are also newer here, and newer houses fetch more, so the size-only line gives size credit for part of the age effect. Leaving out a related variable like this is

omitted-variable bias.

- R^2 rises from 0.958 to 0.982 when age is added: the extra variable explains additional variability.

The full output for the multivariate model, in the style of the coefficient table of Topic 05:

	Coefficient	Standard Error	t Stat	P-value
Intercept	62.90	10.90	5.8	< 0.001
Size	1.96	0.05	43.1	< 0.001
Age	−1.72	0.14	−12.6	< 0.001

Both size and age are significant (every p-value is below 0.001), so each earns its place in the model. The F-test for the whole model has $p < 0.001$ as well. Adjusted $R^2 = 0.982$.

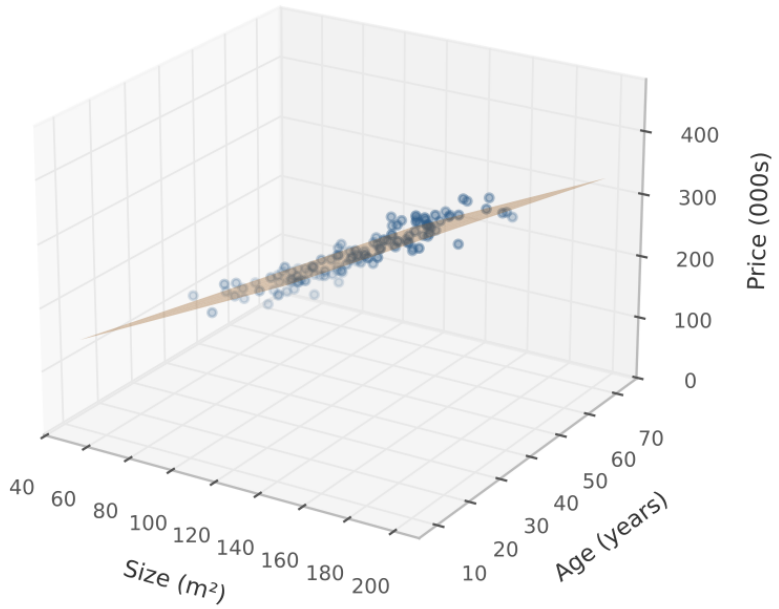


Figure 1. With two predictors the fitted model is a plane. The dots are the simulated houses.

COMMON MISTAKE **Simulated data**

The numbers in this example come from a computer simulation with a fixed random seed. They illustrate the idea; they are not real house prices.

4. How the coefficients are found

OLS minimises SSE exactly as before. In matrix form, with a column of ones for the intercept and one column per variable, the solution is:

$$\mathbf{b} = (X^T X)^{-1} X^T \mathbf{y}$$

You never calculate this by hand. Software does it (Topic 04) and prints the same three tables as before (Topic 05): the model summary, an ANOVA table whose F-test now asks whether **at least one** slope is non-zero, and one row per coefficient with its own t-test and p-value.

5. Conditions for a trustworthy model

Condition	Meaning
Linearity	y is a linear function of the x variables (check with residual plots)
Independence	Observations do not influence each other
Constant variance	The scatter of residuals is similar for all fitted values
Normal errors	Residuals are roughly normal (matters most in small samples; Topic 06 of the inferential course)
No perfect multicollinearity	No x is an exact combination of the others. If two x variables move almost together (say height in cm and in inches), the model cannot tell their effects apart, so their coefficients become unstable with huge standard errors

These conditions are not discussed in the lecture, but results about p-values and intervals depend on them.

> **COMMON MISTAKE** **More variables is not automatically better**
> Each variable you add can only raise R^2 or leave it unchanged, even if it is useless. Adding variables carelessly overfits the sample. Topic 07 shows how to compare models fairly.

6. In Python

```
import numpy as np
import statsmodels.api as sm

# columns: size, age; target: price (simulated data)
X = sm.add_constant(np.column_stack([size, age]))
model = sm.OLS(price, X).fit()
print(model.params)        # b0, b_size, b_age
print(model.summary())     # same three tables as Topic 05

# prediction for a 120 m2 house that is 10 years old
print(model.predict([[1, 120, 10]]))
```

> **SUMMARY**
> - Multivariate regression: $\hat{y} = b_0 + b_1x_1 + \ldots + b_kx_k$; population model adds the error ε.
> - Each coefficient is an effect with the other variables held constant.
> - Geometry: line ($k = 1$), plane ($k = 2$), hyperplane ($k \geq 3$).
> - Omitting a related variable can bias the remaining coefficients.
> - OLS, R^2, ANOVA and coefficient tables work as in simple regression; the F-test is now about all slopes together.
> - R^2 never falls when variables are added, so compare models with the adjusted R^2 (next topic).

Practice questions

Try each question before reading the answer underneath it.

Q1. In $\hat{y} = 10 + 2x_1 - 3x_2$, interpret -3.

Answer. Holding x_1 constant, a one-unit increase in x_2 lowers the predicted y by 3.

Q2. Predict y for $x_1 = 4$, $x_2 = 1$ in the equation above.

Answer. $\hat{y} = 10 + 2 \times 4 - 3 \times 1 = 15$.

Q3. Why can the coefficient on size change when age is added to the model?

Answer. Size and age are correlated, so the simple regression's size coefficient absorbs part of the age effect; adding age separates them.

Q4. What does the line 'best fit' become when there are two predictors?

Answer. A plane (and a hyperplane for three or more predictors).

Q5. Is an R^2 of 0.45 from a model with 10 predictors better than 0.44 from a model with 2? Why or why not?

Answer. Not necessarily. R^2 rises with extra variables even when they are useless; compare adjusted R^2 instead.